Assignment 4

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Packet Tracer - TCP and UDP Communications

Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following task to reduce the amount of network traffic viewed in the simulation.

a. Click MultiServer and click the Desktop tab > Command Prompt.

b. Enter the ping -n 1 192.168.1.255 command. You are pinging the broadcast address for the client LAN. The command option will send only one ping request rather than the usual four. This will take a few seconds as every device on the network responds to the ping request from MultiServer.

```
C:\>ping -n 1 192.168.1.255

Pinging 192.168.1.255 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.3: bytes=32 time=15ms TTL=128
Reply from 192.168.1.4: bytes=32 time=15ms TTL=128

Ping statistics for 192.168.1.255:
    Packets: Sent = 1, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 15ms, Average = 10ms
```

Figure 1. Ping -n 1 192.168.1.255

c. Close the MultiServer window.

Step 2: Generate web (HTTP) traffic.

a. Switch to Simulation mode.

b. Click HTTP Client and open the Web Browser from the desktop.

c. In the URL field, enter 192.168.1.254 and click Go. Envelopes (PDUs) will appear in the topology window.

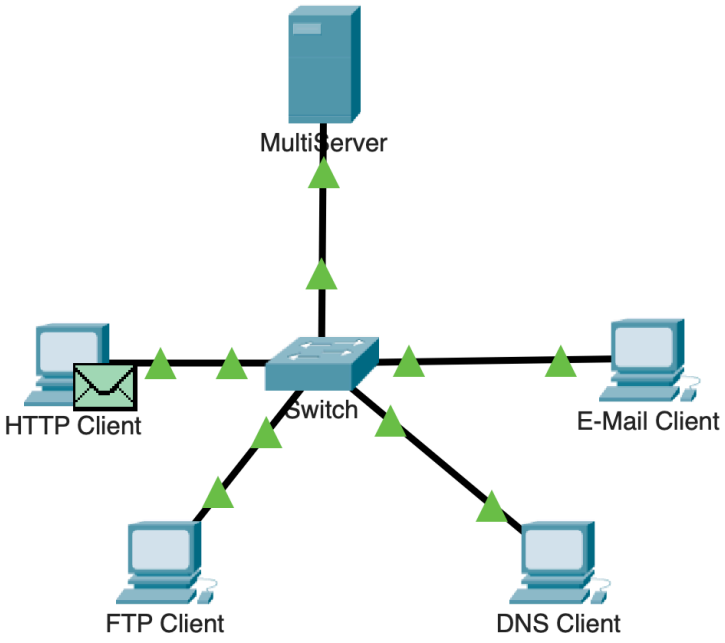

Figure 2. HTTP Client's envelope

d. Minimize, but do not close, the HTTP Client configuration window.

Step 3: Generate FTP traffic.

a. Click FTP Client and open the Command Prompt from the desktop

b. Enter the ftp 192.168.1.254 command. PDUs will appear in the simulation window.

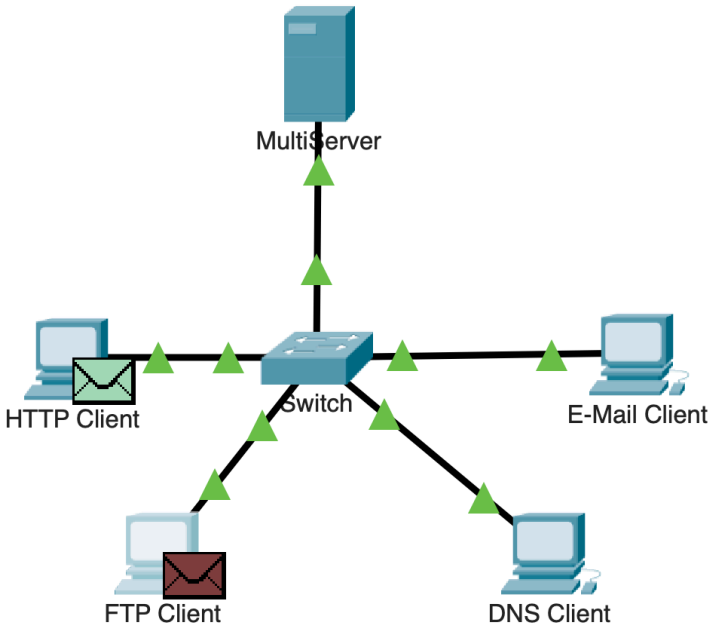

Figure 3. FPT Client's envelope

c. Minimize, but do not close, the FTP Client configuration window.

Step 4: Generate DNS traffic.

a. Click DNS Client and open the Command Prompt.

b. Enter the nslookup multiserver.pt.ptu command. A PDU will appear in the simulation window.

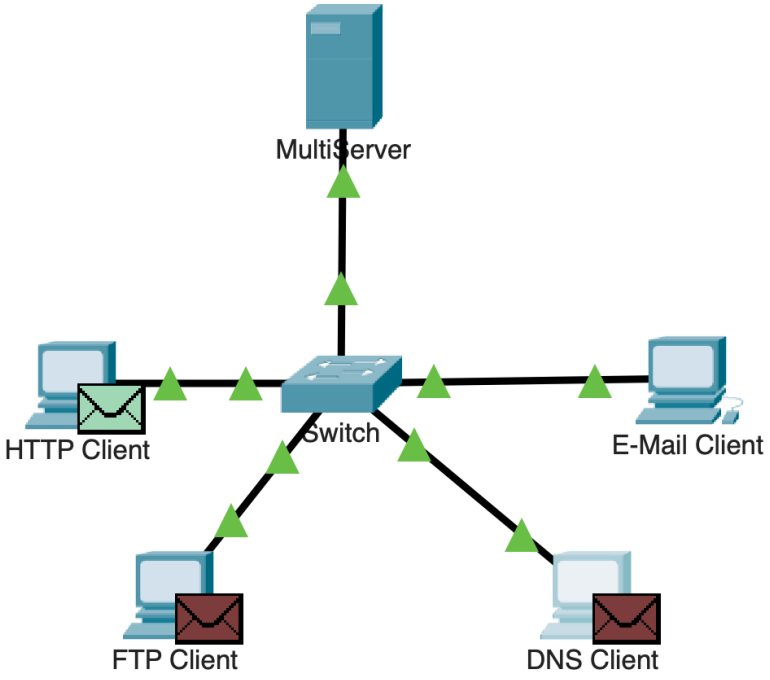

Figure 4. DSN Client's envelop

c. Minimize, but do not close, the DNS Client configuration window.

Step 5: Generate Email traffic.

a. Click E-Mail Client and open the E-Mail tool from the Desktop.

b. Click Compose and enter the following information:

1) To: user@multiserver.pt.ptu

2) Subject: personalize the subject line

3) E-Mail Body: personalize the Email

c. Click Send.

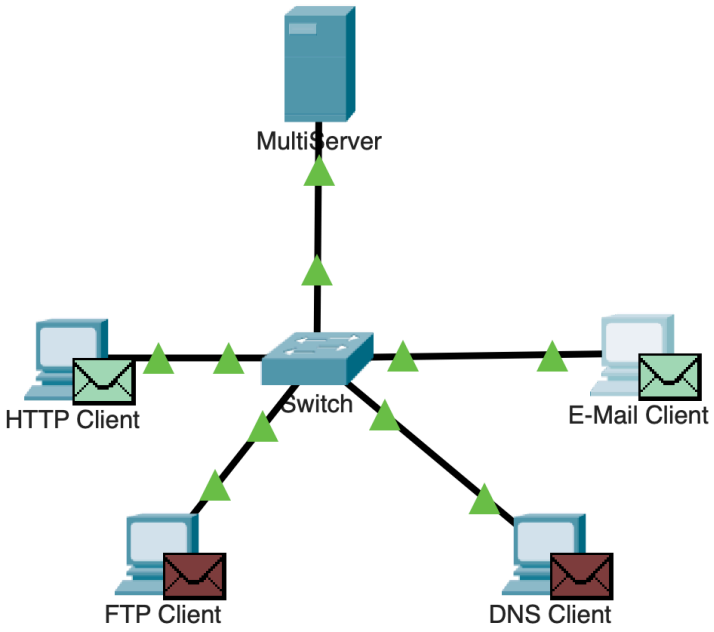

Figure 5. E-Mail's envelop

d. Minimize, but do not close, the E-Mail Client configuration window.

Step 6: Verify that the traffic is generated and ready for simulation.

There should now be PDU entries in the simulation panel for each of the client computers.

Step 7: Examine multiplexing as the traffic crosses the network.

You will now use the Capture/Forward button in the Simulation Panel to observe the different protocols travelling on the network.

Note: The Capture/Forward button ‘ >| ‘ is a small arrow pointing to the right with a vertical bar next to it.

a. Click Capture/Forward once. All of the PDUs travel to the switch.

b. Click Capture/Forward six times and watch the PDUs from the different hosts as they travel on the network. Note that only one PDU can cross a wire in each direction at any given time.

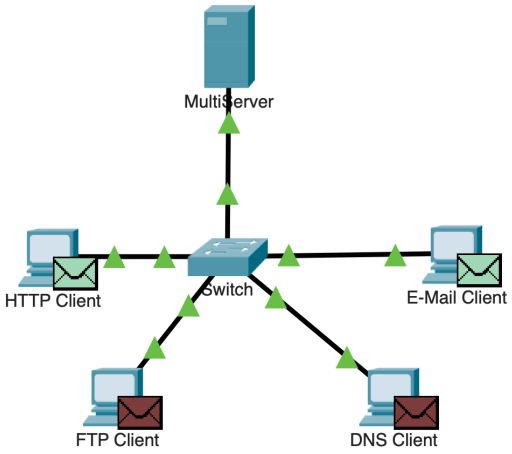

Figure 7.1. PDUs are generated at the source devices and are ready for transmission.

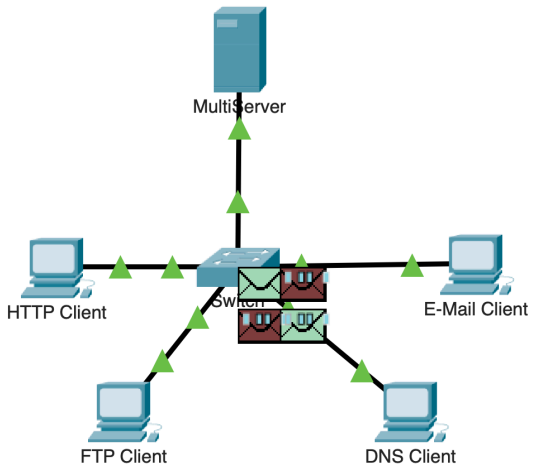

Figure 7.2. The packets travel from the client devices and converge at the central Switch for routing.

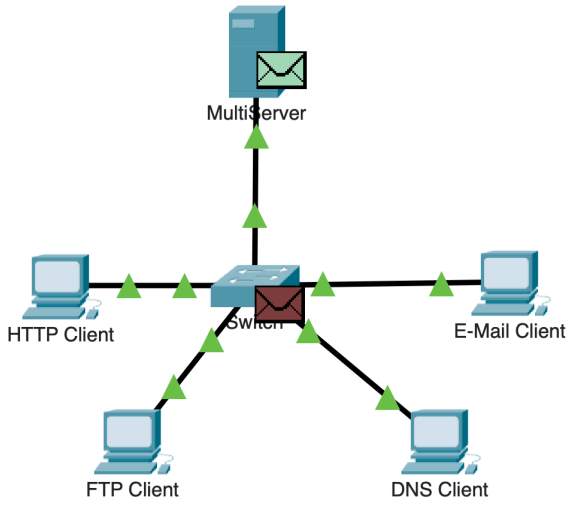

Figure 7.3. The Switch forwards the first green PDU to the MultiServer while holding a brown PDU in the buffer.

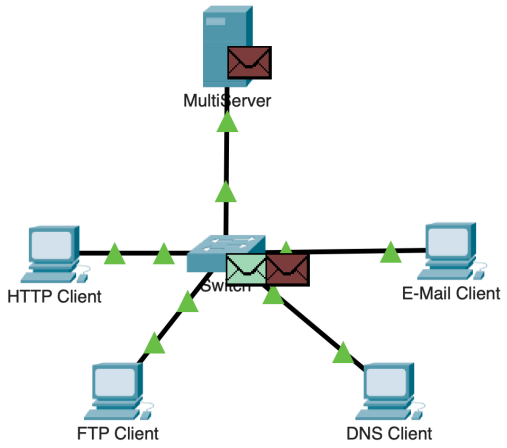

Figure 7.4. The MultiServer processes a received packet while the Switch manages two other queued PDUs.

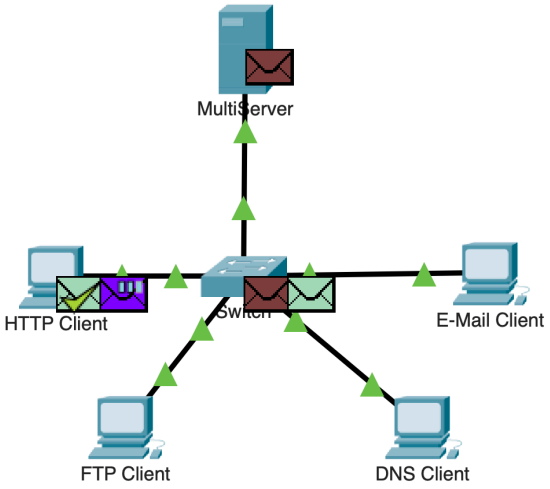

Figure 7.5. The HTTP Client receives a successful acknowledgment (checkmark) and generates a new purple application-layer packet.

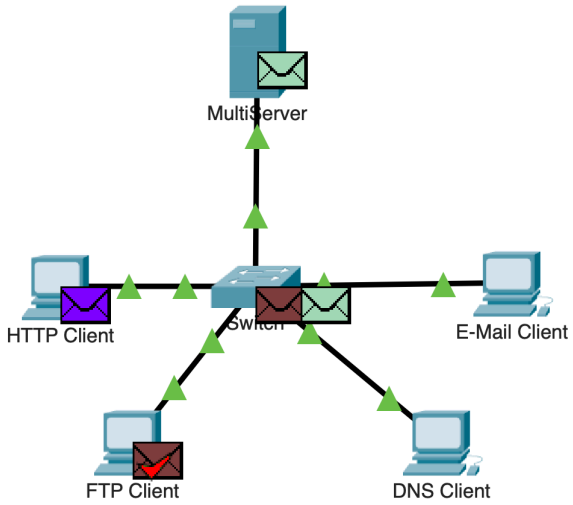

Figure 7.6. The FTP Client receives a successful response, while the MultiServer accepts a new packet and the HTTP Client holds its request.

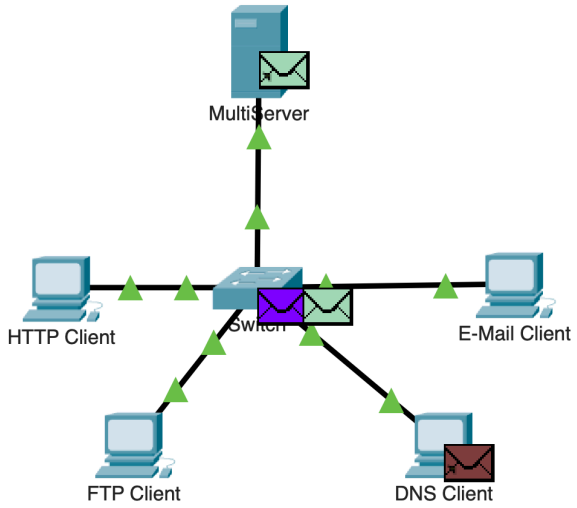

Figure 7.7. The purple application packet from the HTTP Client moves to the Switch to be routed to the server.

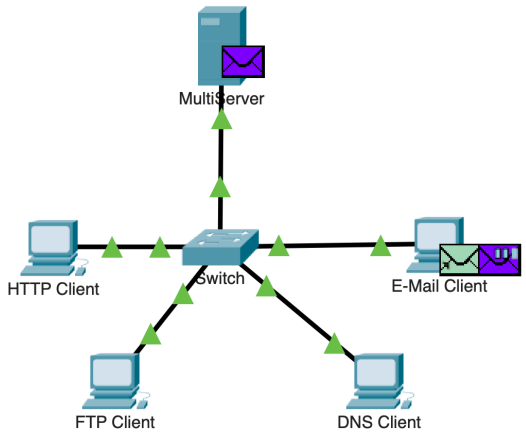

Figure 7.8. The application packet successfully reaches the MultiServer, and the E-Mail Client receives its response packets.

Questions:

What is this called?

This is called Multiplexing

A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors?

The different protocols and types of traffic

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine HTTP traffic as the clients communicate with the server.

a. Click Reset Simulation.

b. Filter the traffic that is currently displayed to only HTTP and TCP PDUs. To filter the traffic that is currently displayed:

1) Click Edit Filters and toggle the Show All/None button.

2) Select HTTP and TCP. Click the red “x” in the upper right-hand corner of the Edit Filters box to close it. Visible Events should now display only HTTP and TCP PDUs.

c. Open the browser on HTTP Client and enter 192.168.1.254 in the URL field. Click Go to connect to the server over HTTP. Minimize the HTTP Client window.

d. Click Capture/Forward until you see a PDU appear for HTTP. Note that the color of the envelope in the topology window matches the color code for the HTTP PDU in the Simulation Panel.

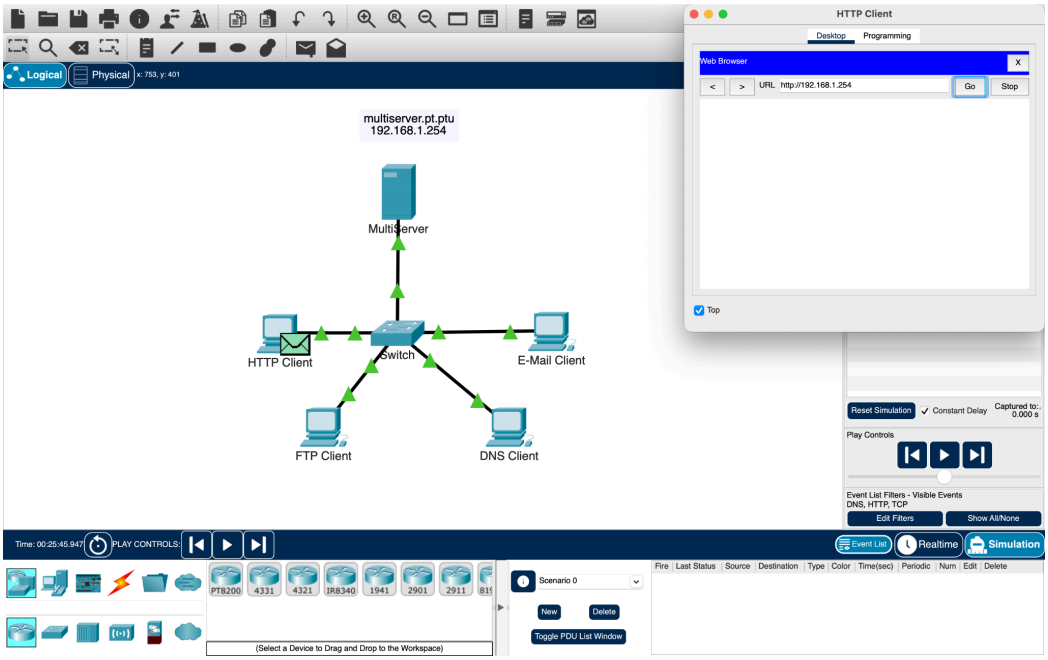

Figure 8.1. The user initiates a web request to the MultiServer IP address using the HTTP Client's web browser.

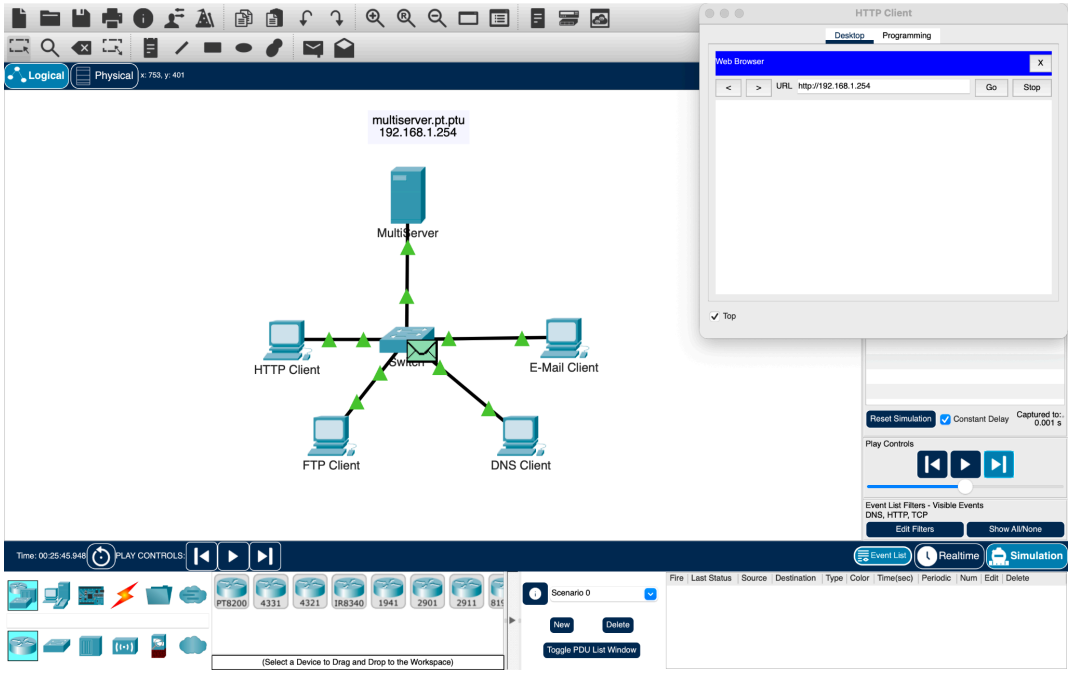

Figure 8.2. The packet is transmitted from the HTTP Client to the Central Switch.

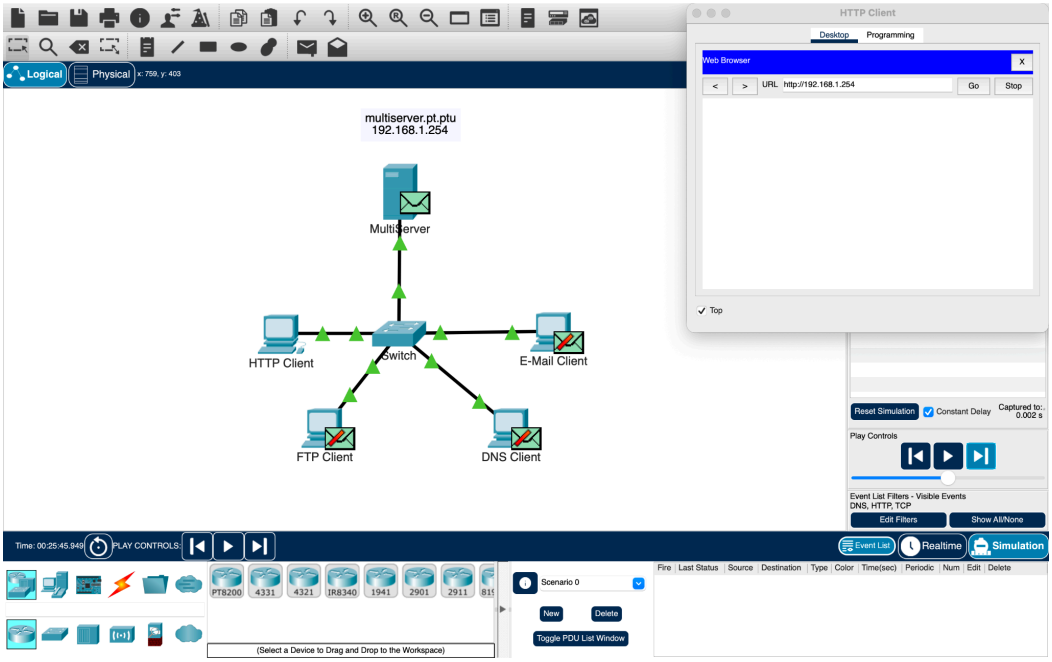

Figure 8.3. The Switch broadcasts/forwards the packet to all connected nodes; it is accepted by the MultiServer but rejected by other clients.

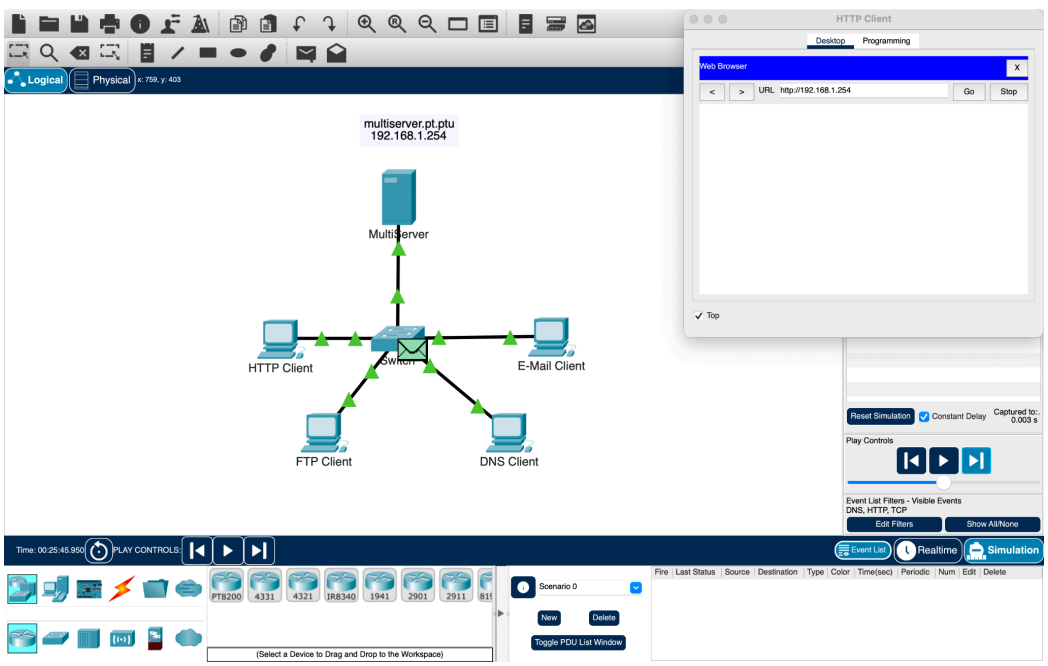

Figure 8.4. The MultiServer processes the request and prepares a response packet to send back to the switch.

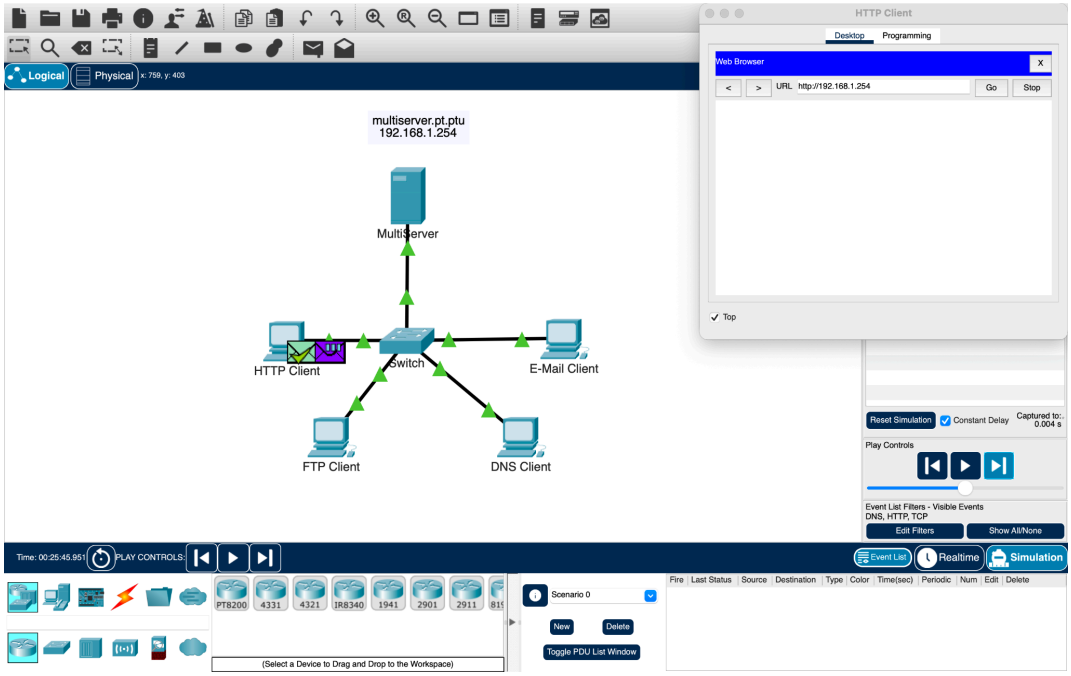

Figure 8.5. The MultiServer sends the response packet back to the switch, while a TCP acknowledgment (purple envelope) appears at the HTTP Client.

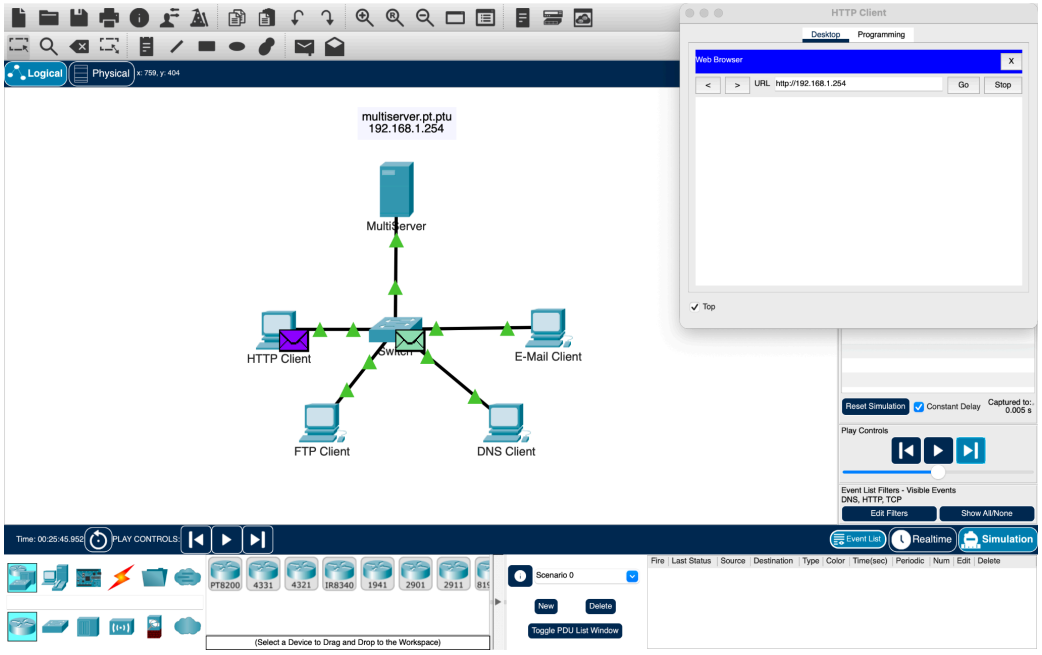

Figure 8.6. The HTTP Client receives the green response packet while simultaneously sending a TCP synchronization/acknowledgment back to the network.

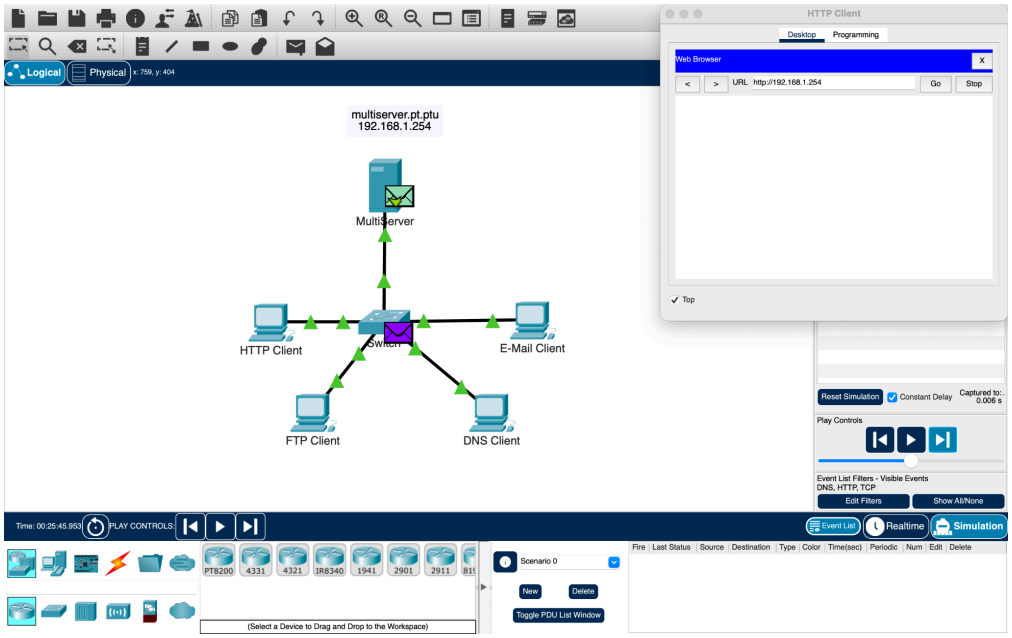

Figure 8.7. Data packets and acknowledgments continue to transit through the Central Switch to manage the session.

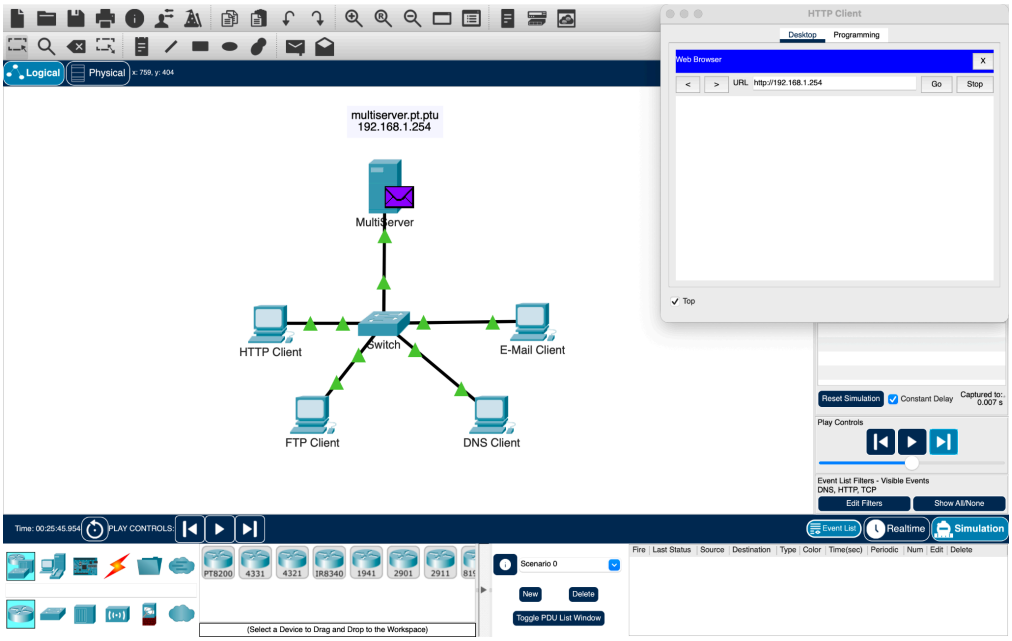

Figure 8.8. The MultiServer receives the final TCP handshake/acknowledgment from the client to stabilize the connection.

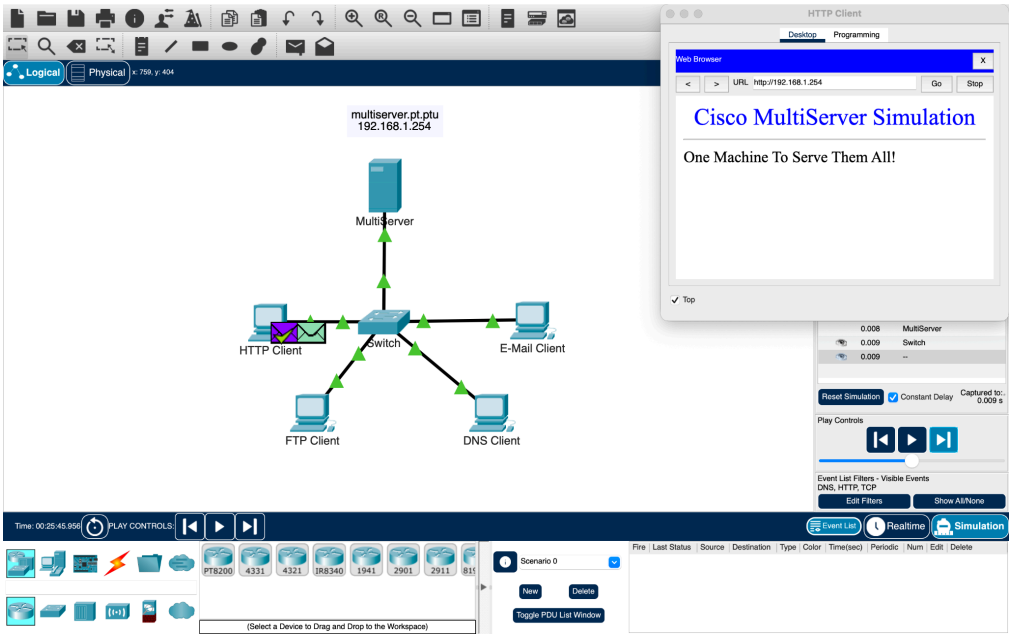

Figure 8.9. The transmission continues as the Switch routes the remaining TCP control packets between the server and the client.

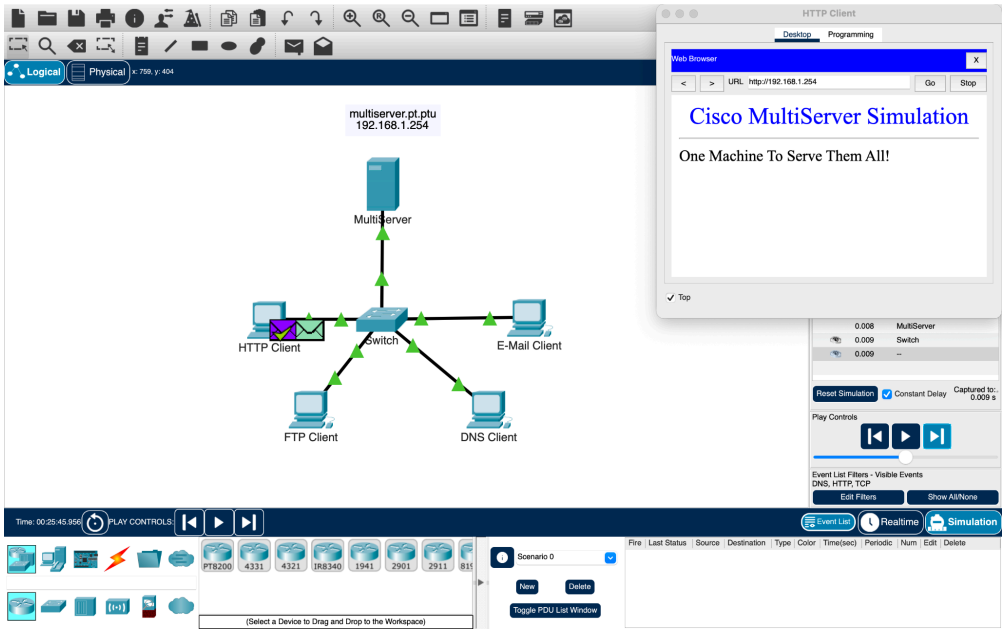

Figure 8.10. The web page is successfully loaded on the HTTP Client, displaying the server's content after the data transmission is complete.

Question:

Why did it take so long for the HTTP PDU to appear?

TCP must first establish a connection using the three-way handshake (SYN, SYN-ACK, ACK) before HTTP data can be transmitted.

e. Click the PDU envelope to show the PDU details. Click the Outbound PDU Details tab and scroll down to the second to the last section.

Questions:

What is the section labeled?

The section is labeled Transmission Control Protocol (TCP).

Are these communications considered to be reliable?

Yes, TCP ensures data delivery using acknowledgements and sequencing.

Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.

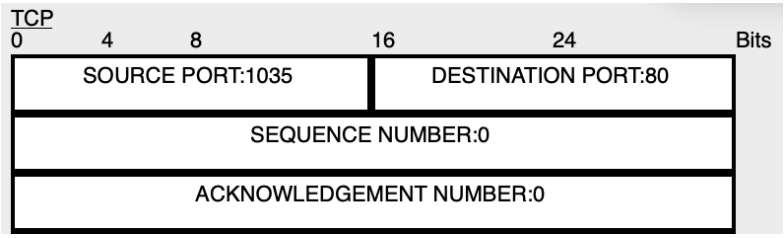

Figure 9. the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.

f. Look at the value in the Flags field, which is located next to the Window field. The values to the right of the “b” represent the TCP flags that are set for this stage of the data conversation. Each of the six places corresponds to a flag. The presence of a “1” in any place indicates that the flag is set. More than one flag can be set at a time. The values for the flags are shown below.

Flag Place	**6**	**5**	**4**	**3**	**2**	**1**
Value	URG	ACK	PSH	RST	SYN	FIN

Which TCP flags are set in this PDU?

The SYN flag is set.

g. Close the PDU and click Capture/Forward until a PDU with a checkmark returns to the HTTP Client.

h. Click the PDU envelope and select Inbound PDU Details.

Question:

How are the port and sequence numbers different than before?

The source and destination ports are reversed, and the sequence and acknowledgement numbers have changed.

i. Click the HTTP PDU which HTTP Client has prepared to send to MultiServer. This is the beginning of the HTTP communication. Click this second PDU envelope and select Outbound PDU Details.

Question:

What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

The TCP section now includes data transfer information such as sequence and acknowledgement numbers. The ports remain the same, but the sequence number has increased and the acknowledgement number reflects previous communication.

j. Reset the simulation.

Step 2: Examine FTP traffic as the clients communicate with the server.

a. Open the command prompt on the FTP Client desktop. Initiate an FTP connection by entering ftp 192.168.1.254.

b. In the Simulation Panel, change Edit Filters to display only FTP and TCP.

c. Click Capture/Forward. Click the second PDU envelope to open it.

Click the Outbound PDU Details tab and scroll down to the TCP section.

Question:

Are these communications considered to be reliable?

Yes, these communications are considered reliable.

d. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.

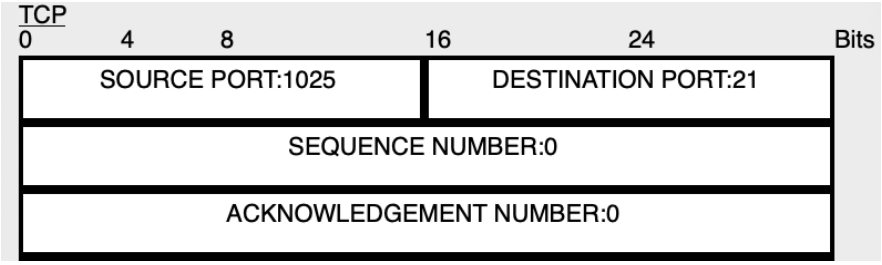

Figure 10. TCP section in Outbound PDU Details

Question:

What is the value in the flag field?

The value in the flag field is SYN.

e. Close the PDU and click Capture/Forward until a PDU returns to the FTP Client with a checkmark.

f. Click the PDU envelope and select Inbound PDU Details.

Question:

How are the port and sequence numbers different than before?

The source and destination ports are reversed, and the sequence and acknowledgement numbers have changed compared to the previous PDU.

g. Click the Outbound PDU Details tab.

Question:

How are the port and sequence numbers different from the previous results?

The source and destination ports remain the same, but the sequence number has increased, and the acknowledgement number reflects receipt of the previous data.

h. Close the PDU and click Capture/Forward until a second PDU returns to the FTP Client. The PDU is a different color.

i. Open the PDU and select Inbound PDU Details. Scroll down past the TCP section.

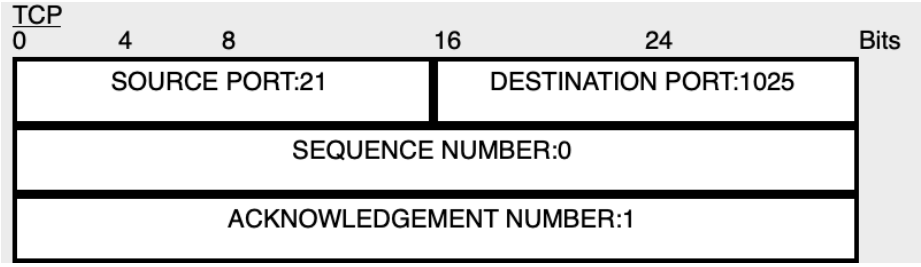

Figure 11. TCP section in Inbound PDU Details

Question:

What is the message from the server?

220- Welcome to PT Ftp server Username:

j. Click Reset Simulation.

Step 3: Examine DNS traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to create DNS traffic.

b. In the Simulation Panel, change Edit Filters to display only DNS and UDP.

c. Click the PDU envelope to open it.

d. Look at the OSI Model details for the outbound PDU.

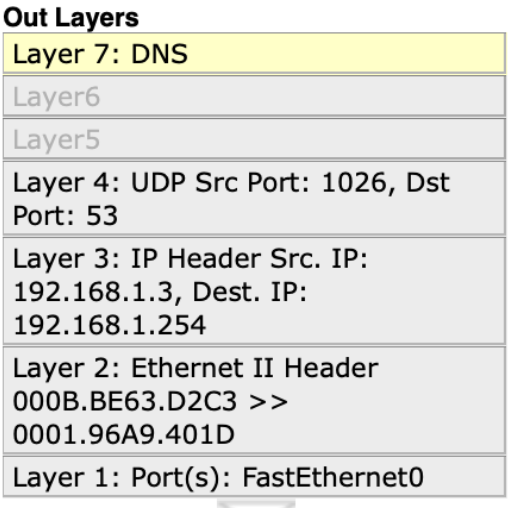

Figure 12. Out Layer in OSI Model

Question:

What is the Layer 4 protocol?

The Layer 4 protocol is UDP.

Are these communications considered to be reliable?

No, these communications are not reliable.

e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the SRC PORT and DEST PORT values.

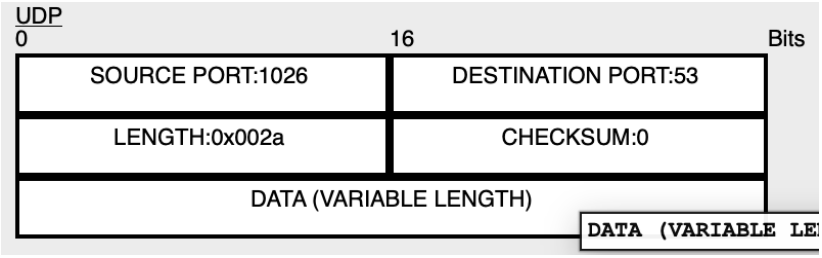

Figure 13. UDP section in Outbound PDU details of DSN Client

Question:

Why are there no sequence and acknowledgement numbers?

There are no sequence and acknowledgement numbers because UDP is a connectionless protocol.

f. Close the PDU and click Capture/Forward until a PDU with a check mark returns to the DNS Client.

g. Click the PDU envelope and select Inbound PDU Details.

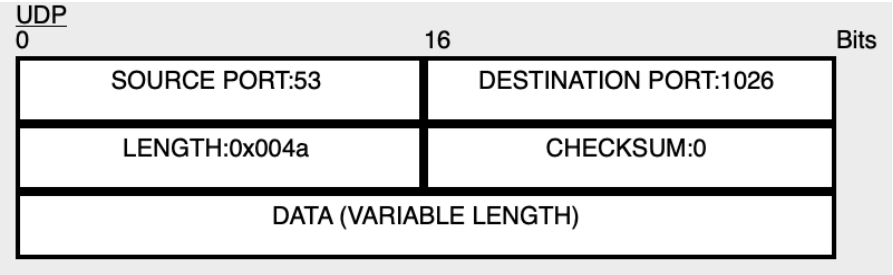

Figure 14. UDP section in Inbound PDU details of DSN Client

Question:

How are the port and sequence numbers different than before?

The source and destination ports are reversed, and there are no sequence or acknowledgement numbers.

What is the last section of the PDU called? What is the IP address for the name multiserver.pt.ptu?

The last section of the PDU is called DNS and The IP address for the name multiserver.pt.ptu is 192.168.1.254.

h. Click Reset Simulation.

Step 4: Examine email traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to send an email to user@multiserver.pt.ptu.

b. In the Simulation Panel, change Edit Filters to display only POP3, SMTP and TCP.

c. Click the first PDU envelope to open it.

d. Click the Outbound PDU Details tab and scroll down to the last section.

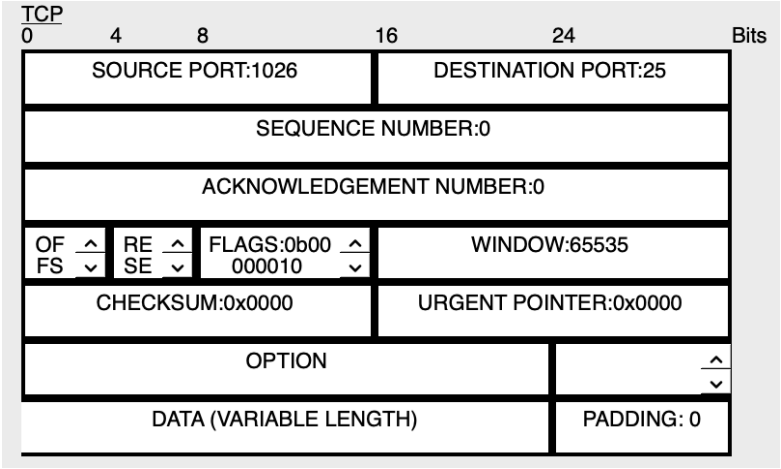

Figure 15. TCP section in Outbound PDU Details of E-Mail Client

Questions:

What transport layer protocol does email traffic use?

Email traffic uses TCP as its transport layer protocol.

Are these communications considered to be reliable?

Yes, these communications are considered reliable.

e. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values. What is the flag field value?

The flag field value is SYN.

f. Close the PDU and click Capture/Forward until a PDU returns to the E-Mail Client with a checkmark.

g. Click the TCP PDU envelope and select Inbound PDU Details.

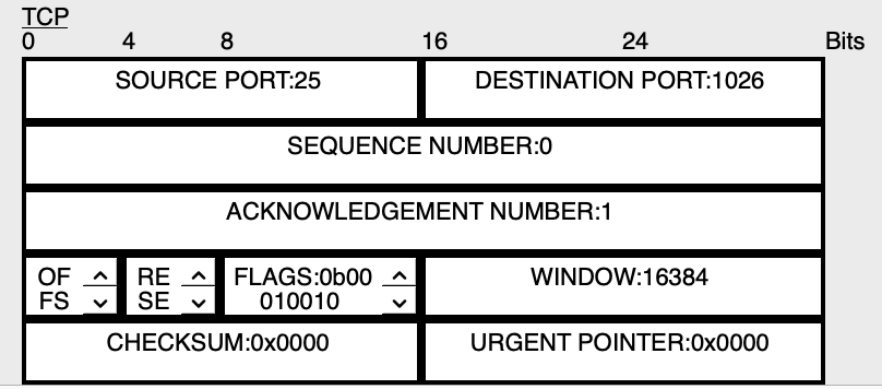

Figure 16. TCP section in Inbound PDU Details of E-Mail Client

Question:

How are the port and sequence numbers different than before?

The source and destination ports are reversed, and the sequence and acknowledgement numbers have changed.

h. Click the Outbound PDU Details tab.

Question:

How are the port and sequence numbers different from the previous two results?

The ports remain the same, but the sequence number has increased and the acknowledgement number is updated.

i. There is a second PDU of a different color that E-Mail Client has prepared to send to MultiServer. This is the beginning of the email communication. Click this second PDU envelope and select Outbound PDU Details.

Questions:

How are the port and sequence numbers different from the previous two PDUs?

A new sequence number is used because email data transmission begins.

What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

SMTP protocol is associated with TCP port 25 and POP3 protocol is associated with TCP prot 110.